

Forcing-sterility of the realizable unit-distance lineage and a codegree obstruction to $\chi \geq 6$

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July 2026

Abstract

The chromatic number of the plane, $\chi(\mathbb{R}^2)$, is known only to lie in $[5, 7]$. The lower bound $\chi(\mathbb{R}^2) \geq 5$ rests on finite unit-distance graphs (UDGs): de Grey (2018) exhibited the first such graph with $\chi \geq 5$, and the Polymath16 project reduced the order to roughly 510 vertices. A natural route to $\chi(\mathbb{R}^2) \geq 6$ would locate, somewhere in this realizable lineage of $\chi = 5$ UDGs, a non-adjacent vertex pair forced to distinct colors in every proper 5-coloring (a *clamp*); realizing such a pair lifts the chromatic number to 6. This note records a negative structural result: the realizable lineage contains no such forcing, and provably so. We recall an elementary lemma (the *Essential-Pair Lemma*, essentially known) which implies that a vertex-critical 5-chromatic graph cannot host any forced non-adjacent pair. We confirm the consequence by an exhaustive computational census: all 1,955,948 non-adjacent pairs across the nine $\chi = 5$ lineage graphs are free, with zero forced pairs of either kind. Since the lineage is vertex-critical, this is a one-line corollary of the Lemma, and the deeper point is conceptual: the lineage was produced by SAT-driven minimization, which drives graphs toward vertex-criticality, and criticality deletes exactly the vertices that could host forcing. The lineage is forcing-sterile *by construction*. We then frame the search for the missing $\chi \geq 6$ object through a codegree obstruction: every planar UDG is $K_{2,3}$ -free, so its edge count is bounded by $m \leq n(1 + \sqrt{8n-7})/4$; combined with the Kostochka-Yancey lower bound for 6-critical graphs and the vertex Folkman number $F_v(2, 2, 2, 2, 2; 4) = 16$, this excludes every small dense Folkman-type host. The missing object must lie in the class of graphs that are simultaneously K_4 -free and $K_{2,3}$ -free. For that class we prove an exhaustive negative: over the full edge window between the Kostochka-Yancey floor and the codegree ceiling, the class contains no member with chromatic number at least 6 on at most 17 vertices, so the smallest such graph, if one exists, has at least 18 vertices; density heuristics place it closer to order 26 or beyond. Through order 16 this is a custom-pruned **geng** enumeration, confirmed graph-for-graph by an independent second enumerator (all 11,315 candidates in the order-16 window are 5-colorable). At order 17, where enumeration costs more than 80 core-days, we instead decide the question without enumerating: a chromatic-number propagator inside a symmetry-breaking SAT solver turns the two-quantifier question into a single unsatisfiability proof per edge window cell, and the order-17 window closes in under one core-hour. We claim no new bound on $\chi(\mathbb{R}^2)$.

1 Introduction

The Hadwiger-Nelson problem asks for the chromatic number $\chi(\mathbb{R}^2)$ of the *unit-distance graph* on the Euclidean plane: vertices are the points of $\mathbb{R}^2$, and two points are adjacent iff they are at distance exactly 1. The classical bounds are

$$5 \leq \chi(\mathbb{R}^2) \leq 7,$$

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and have resisted improvement for three quarters of a century. The upper bound $\chi(\mathbb{R}^2) \leq 7$ is the Isbell hexagonal coloring (tile the plane with regular hexagons of diameter slightly less than 1 and color in a repeating 7-pattern). The lower bound $\chi(\mathbb{R}^2) \geq 5$ is recent and computational: de Grey [2] constructed a finite unit-distance graph on 1581 vertices with $\chi \geq 5$, verified by a SAT proof, and the Polymath16 project [11] subsequently reduced the order of such graphs to roughly 510 vertices [10]. We refer to this family of explicitly realized $\chi = 5$ UDGs as the *realizable lineage*. For background on the problem and its history see Soifer [12].

The natural finitary route to $\chi(\mathbb{R}^2) \geq 6$ is to find a finite UDG with $\chi \geq 6$. One concrete mechanism for manufacturing such a graph out of a $\chi = 5$ graph is a *clamp*.

Definition 1 (Clamp). Let G be a 5-chromatic graph and (s, t) a non-adjacent pair of vertices. We call (s, t) a *clamp* if every proper 5-coloring of G assigns $c(s) \neq c(t)$.

If a clamp (s, t) in a UDG can be *realized*, that is, if the geometry permits placing a vertex at unit distance from both s and t while preserving the unit-distance structure (so that an edge effectively appears across the forced-different pair), the chromatic number is lifted to 6. The motivating question for this note is therefore:

Does the realizable lineage of $\chi = 5$ UDGs contain a clamp (or any other forced non-adjacent pair) that could be exploited to reach $\chi \geq 6$?

The answer of this note is **no, and provably so**. We make this precise through one elementary lemma (Section 3), confirm it with an exhaustive computational census (Section 4), explain it structurally by way of vertex-criticality (Section 5), and then reframe the search for the genuinely missing object through a codegree obstruction (Section 6), and settle the small-order cases of the resulting class outright (Section 7). Section 8 synthesizes the picture and states the target object honestly.

Scope and honesty. This is a negative, explanatory note. We move no bound on $\chi(\mathbb{R}^2)$. The contribution is fourfold: (a) the observation that the realizable lineage is forcing-sterile *by construction*, obtained by applying a known elementary lemma to its vertex-criticality; (b) an exhaustive computational census confirming it across the entire lineage; (c) a codegree framing that locates the missing $\chi \geq 6$ object outside the reach of both the realizable lineage and the small dense graphs one would naively manufacture; and (d) an exhaustive negative showing that the class the codegree framing isolates contains no $\chi \geq 6$ member on at most 17 vertices, so the missing object has order at least 18. None of the four ingredients is individually deep; the synthesis is the point.

2 Preliminaries

All graphs are finite, simple, and undirected. For a graph G we write $V(G)$ for its vertex set, $n = |V(G)|$, $m = |E(G)|$, and $\chi(G)$ for the chromatic number. A graph is *5-chromatic* if $\chi(G) = 5$. A proper k -coloring is a map $c : V(G) \rightarrow \{1, \dots, k\}$ with $c(u) \neq c(v)$ whenever $uv \in E(G)$.

Definition 2 (Vertex-critical). A graph G is *vertex-critical* (with respect to its chromatic number) if $\chi(G - v) < \chi(G)$ for every $v \in V(G)$. A 5-chromatic graph is vertex-critical iff $\chi(G - v) \leq 4$ for every vertex v .

Definition 3 (Forcing of a non-adjacent pair). Let G be a 5-chromatic graph and let (s, t) be a pair of distinct, *non-adjacent* vertices. We classify (s, t) as follows.

- **FORCED-SAME:** every proper 5-coloring of G has $c(s) = c(t)$.
- **FORCED-DIFFERENT:** every proper 5-coloring of G has $c(s) \neq c(t)$.
- **FORCED:** either **FORCED-SAME** or **FORCED-DIFFERENT** holds.
- **FREE:** neither holds, i.e. there exist a proper 5-coloring with $c(s) = c(t)$ and a (possibly different) proper 5-coloring with $c(s) \neq c(t)$.

A clamp (Definition 1) is exactly a forced-different non-adjacent pair. A forced-same pair is no weaker as a route to $\chi \geq 6$: it is *already* a $\chi \geq 6$ certificate, since if every proper 5-coloring of G assigns $c(s) = c(t)$, then adding the edge st (when the geometry realizes it as a unit distance) yields a graph with no proper 5-coloring at all, i.e. $\chi \geq 6$. For the purposes of this note the relevant dichotomy is simply *forced* versus *free*: a single forced pair anywhere in the lineage would be a candidate $\chi \geq 6$ ingredient.

Deciding forcing by SAT. Each forcing question reduces to two k -colorability decisions, both decidable by a standard k -coloring SAT encoding.

- **FORCED-DIFFERENT** holds for (s, t) iff the graph $G/\{s, t\}$, obtained by *identifying* (merging) s and t , is not 5-colorable. Merging forces $c(s) = c(t)$; if even that constrained instance is 5-UNSAT, then no proper 5-coloring has $c(s) = c(t)$, so every proper 5-coloring has $c(s) \neq c(t)$.
- **FORCED-SAME** holds for (s, t) iff the graph $G + st$, obtained by *adding* the edge st , is not 5-colorable. Adding the edge forbids $c(s) = c(t)$; if that instance is 5-UNSAT, then no proper 5-coloring has $c(s) \neq c(t)$, so every proper 5-coloring has $c(s) = c(t)$.
- Otherwise (s, t) is **FREE**.

Both tests are decisive: a 5-UNSAT verdict from a complete solver is a proof, and a satisfying assignment is an explicit certificate of freeness on that side.

Control objects. Throughout we keep in mind the structural controls of the program. The unit-distance graph on $\mathbb{Q}^2$ has chromatic number 2 (Woodall [14]); any purely combinatorial argument for $\chi(\mathbb{R}^2) \geq 5$ must use the topology of $\mathbb{R}$ in a way that fails on $\mathbb{Q}^2$. The codegree obstruction of Section 6, and the class enumeration of Section 7 built on it, are genuinely Euclidean: they use two-circle rigidity, which fails under the sup-norm (where unit “circles” are squares and can share a segment). The Essential-Pair Lemma of Section 3, by contrast, is pure graph theory and claims no bound anywhere.

3 The Essential-Pair Lemma

The following lemma is elementary and essentially known. Its corollary, that a vertex-critical k -chromatic graph admits no forced non-adjacent pair, is implicit in standard critical-graph theory (Dirac and Gallai; see Jensen and Toft [4]). It also appears explicitly as Theorem 3.17 (“there are no implicit relations in critical graphs”) of Martin [8], where forced-different and forced-same pairs are termed *implicit-edges* and *implicit-identities*; the single-deletion form we state is a one-line sharpening of his Theorems 3.2 and 3.17. We state and prove it for self-containedness and make no priority claim; the proof uses only the definition of a proper coloring.

Lemma 1 (Essential-Pair Lemma). *Let G be a 5-chromatic graph and let (s, t) be a non-adjacent pair that is forced (forced-same or forced-different). Then*

$$\chi(G - s) \geq 5 \quad \text{and} \quad \chi(G - t) \geq 5.$$

Proof. We prove $\chi(G - s) \geq 5$; the statement for t follows by symmetry.

Suppose toward a contradiction that $\chi(G - s) \leq 4$. Fix a proper 4-coloring of $G - s$ using colors $\{1, 2, 3, 4\}$. We extend it to G in two different ways.

The “different” pattern. Give s the new color 5. Every neighbor of s lies in $G - s$ and so is colored with a color in $\{1, 2, 3, 4\}$; hence no neighbor of s has color 5, and the extension is a proper 5-coloring of G . In it, $c(s) = 5$. Since $t \neq s$ and t is colored with a color in $\{1, 2, 3, 4\}$, we have $c(s) \neq c(t)$. Thus the forced-different pattern is realized.

The “same” pattern. Starting again from the same proper 4-coloring of $G - s$, set $c(s) = 5$ as before, and additionally *recolor* t to 5. Because s and t are **non-adjacent**, every neighbor of t also lies in $G - s$ (a neighbor of t cannot be s), and so is colored in $\{1, 2, 3, 4\}$; recoloring t to 5 therefore creates no monochromatic edge at t . The vertex s likewise has all neighbors in $\{1, 2, 3, 4\}$, so no edge at s is monochromatic either. The result is a proper 5-coloring of G with $c(s) = 5 = c(t)$. Thus the forced-same pattern is realized.

Both the same-colored and the differently-colored patterns occur in proper 5-colorings of G . Hence (s, t) is FREE (Definition 3), contradicting the hypothesis that it is forced. Therefore $\chi(G - s) \geq 5$, and by symmetry $\chi(G - t) \geq 5$. $\square$

Corollary 1. *A vertex-critical 5-chromatic graph contains no forced non-adjacent pair: every non-adjacent pair is free.*

Proof. Let G be vertex-critical and 5-chromatic, so $\chi(G - v) \leq 4$ for every vertex v . If some non-adjacent pair (s, t) were forced, then Lemma 1 would give $\chi(G - s) \geq 5$, contradicting $\chi(G - s) \leq 4$. Hence no non-adjacent pair is forced. $\square$

The content of Corollary 1 is intuitive once stated: a forced pair (s, t) requires that the rest of the graph be “heavy” enough at both s and t to constrain their colors even without an edge between them, and that heaviness is precisely what survives the deletion of s (or t). If the graph is already minimal in the sense that every vertex deletion drops the chromatic number, no such residual heaviness exists.

4 The exhaustive census

We now report the computational confirmation. The realizable lineage of $\chi = 5$ UDGs studied here consists of nine graphs, of orders

$$510, 517, 529, 553, 610, 633, 803, 826, 874,$$

descending from the de Grey / Polymath16 constructions. (An earlier triage table listed three further graphs, on 199, 403, and 721 vertices, which on re-examination have $\chi = 4$; they are excluded here, and their exclusion does not affect the census headline.)

Method. For every non-adjacent pair (s, t) in each of the nine graphs we ran *both* decisive SAT tests of Section 2:

- forced-different is decided by *merging* s and t (identifying the two vertices) and testing whether the merged graph is 5-UNSAT;
- forced-same is decided by *adding* the edge st and testing whether the augmented graph is 5-UNSAT;
- if neither test returns UNSAT, the pair is free.

A per-solve conflict budget was used and tuned so that the number of indeterminate solves was driven to exactly zero; every pair received a decisive classification.

Result. All 1,955,948 non-adjacent pairs across the nine graphs were classified, and **every single one is free**: zero forced-same, zero forced-different, zero indeterminate. Table 1 reports the per-graph counts.

Table 1: Exhaustive forcing census over the realizable $\chi = 5$ lineage. For each graph, every non-adjacent pair was classified by two decisive SAT tests (merge for forced-different, add-edge for forced-same). Every pair is free.

Order n	Non-adjacent pairs	Forced-same	Forced-diff.	Free
510	127,291	0	0	all
517	130,807	0	0	all
529	136,986	0	0	all
553	149,906	0	0	all
610	182,745	0	0	all
633	196,862	0	0	all
803	317,859	0	0	all
826	336,452	0	0	all
874	377,040	0	0	all
Total	1,955,948	0	0	all

Equivalently: in the realizable lineage, every non-adjacent pair admits both a same-colored and a differently-colored proper 5-coloring. All color forcing is *adjacency-local*. A single non-free pair would have been a candidate $\chi \geq 6$ ingredient (a forced-different pair is a clamp directly; a forced-same pair certifies $\chi \geq 6$ by adding the realized edge st). None exists.

5 The criticality explanation

The census of Section 4 is not a coincidence, and it is not even a separate fact. A criticality scan over the same nine graphs (deciding 4-colorability of $G - v$ for every vertex v , totaling on the order of 5,000 solves) found that *all nine lineage graphs are vertex-critical*: deleting any single vertex yields a 4-colorable graph, with zero redundant vertices and zero indeterminate verdicts.

Theorem 1. *Every graph in the realizable $\chi = 5$ lineage is vertex-critical, and therefore (by Corollary 1) contains no forced non-adjacent pair. Consequently the exhaustive census of Section 4 is a one-line corollary of the Essential-Pair Lemma.*

Proof. Vertex-criticality is the computational output of the scan. Given it, Corollary 1 applies to each of the nine graphs and yields that no non-adjacent pair is forced, which is exactly the census conclusion. $\square$

The conceptual heart. Theorem 1 explains the census mechanistically, and the explanation is the central point of this note. The realizable lineage was *produced by SAT-driven minimization*: starting from a large $\chi = 5$ UDG, one repeatedly deletes vertices and edges while preserving $\chi = 5$, driving the graph toward a minimal core. Minimization of this kind drives a graph toward vertex-criticality. But by the Essential-Pair Lemma, criticality deletes *exactly* the vertices that could host a forced pair: a forced pair (s, t) certifies that s and t are chromatically *redundant* (their deletion keeps $\chi \geq 5$), and a vertex-critical graph has no redundant vertices. The production process therefore removes every clamp-capable endpoint before any forcing search is ever run.

The realizable lineage is forcing-sterile by construction. The missing $\chi \geq 6$ ingredient cannot be hiding in it, not because no one searched carefully, but because the very procedure that produced the lineage removed the only vertices that could carry the ingredient.

This has a concrete methodological consequence. A clamp (or any forced pair) can only live at chromatically redundant vertices, so a candidate-generation procedure aimed at $\chi \geq 6$ must *preserve* redundancy: it must **not** minimize toward criticality. Equivalently, any forcing sweep can be pre-filtered by an $O(n)$ criticality scan (keep only redundant vertices) rather than testing all $O(n^2)$ pairs, since forcing can only occur at redundant endpoints. We note one caveat: redundancy is necessary but not sufficient. We have observed non-minimal 5-chromatic graphs (for instance a particular C_6 -closure on 1155 vertices, an unpublished computational observation from the present program) that are forcing-free despite having redundant vertices.

6 A codegree obstruction to small dense hosts

If the missing $\chi \geq 6$ object is not in the realizable lineage (Sections 4-5), the alternative is to manufacture an abstract $\chi = 6$ graph and realize it as a UDG. The natural supply is K_4 -free 6-chromatic graphs (one wants K_4 -freeness because unit distances make K_4 geometrically rigid and hard to realize freely). These exist in abundance and can be made small. The obstruction recorded here is that they are *too dense* to be planar UDGs.

The codegree bound. In the Euclidean plane, two distinct unit circles meet in at most 2 points. Hence any two vertices of a planar UDG have at most 2 common neighbors: **every planar UDG is $K_{2,3}$ -free**. (This is the classical fact underlying the Spencer-Szemerédi-Trotter $O(n^{4/3})$ bound on the number of unit distances among n points [13].)

The $K_{2,3}$ -free condition bounds the edge count. Counting paths of length two through each vertex,

$$\sum_v \binom{d_v}{2} = \#\{\text{paths of length 2}\} \leq 2 \binom{n}{2},$$

because each unordered pair of vertices is the pair of endpoints of at most 2 such paths ($K_{2,3}$ -freeness caps the common-neighbor count at 2). By convexity, $\sum_v \binom{d_v}{2} \geq n \binom{2m/n}{2}$, and solving the resulting quadratic in m gives

$$m \leq \frac{n(1 + \sqrt{8n - 7})}{4}. \tag{1}$$

We call the right-hand side the *codegree ceiling*.

Manufactured hosts overshoot it. Empirically, the smallest reachable K_4 -free 6-chromatic (Folkman-type) hosts are dense, with edge-to-vertex ratio $m/n \approx 4.2$, far above the ceiling of (1). Concretely:

- an 18-vertex K_4 -free 6-critical host has $m = 76$, against a ceiling of $\lfloor 57.17 \rfloor = 57$;
- the classical Mycielskian $M^3(C_5)$ on 47 vertices has 434 vertex pairs of common-neighbor count ≥ 3 , i.e. 434 violations of the $K_{2,3}$ -free condition, with maximum codegree 11.

Such graphs cannot be planar UDGs; the violation is structural (it persists for edge-critical cores, which have no slack to remove), not an artifact of any particular construction heuristic.

The violations are intrinsic to the critical core. One might hope that the $K_{2,3}$ violations of a manufactured host are removable “padding” that a careful sparsification could shed while preserving $\chi = 6$. A top-down probe indicates otherwise. Starting from $M^3(C_5)$ (the 47-vertex $\chi = 6$ Mycielskian above, with 434 violating pairs) and repeatedly deleting edges while holding $\chi = 6$ as a hard constraint (re-establishing 6-chromaticity by a compensating forced-same edge whenever a deletion would drop it) drives the graph down to $m = 131$ edges, essentially the Kostochka-Yancey 6-critical floor $2.8 \cdot 47 - 1.8 = 129.8$. Yet the codegree excess does not fall to zero: the resulting edge-critical, K_4 -free, $\chi = 6$ graph still has 166 vertex pairs of common-neighbor count ≥ 3 . Sparsifying all the way to the critical floor removes none of the $K_{2,3}$ violations; they live entirely in the 6-critical core. For this construction, then, the chromatic obstruction and the codegree obstruction are the *same* substructure: one cannot shed the codegree violations without destroying the very 6-chromaticity they accompany. (This is a heuristic observation about the $M^3(C_5)$ -derived family, obtained by local search, not a theorem about all 6-critical graphs.)

A rigorous core-size floor. The K_4 -free assumption can be dropped for a clean lower bound on the order of the 6-critical core. The Kostochka-Yancey theorem [6] states that every 6-critical graph has

$$m \geq 2.8n - 1.8.$$

Comparing this floor with the codegree ceiling (1), the floor *exceeds* the ceiling for $n \leq 12$ (and they cross at $n = 13$). Hence:

Proposition 1. *Any 6-chromatic unit-distance graph in the plane contains a 6-critical core on at least 13 vertices.*

Proof. A 6-chromatic graph contains a 6-critical subgraph H (delete vertices and edges while $\chi = 6$). As a subgraph of a planar UDG, H is $K_{2,3}$ -free, so $|E(H)| \leq |V(H)|(1 + \sqrt{8|V(H)| - 7})/4$. As a 6-critical graph, $|E(H)| \geq 2.8|V(H)| - 1.8$. For $|V(H)| \leq 12$ the lower bound exceeds the upper bound, a contradiction, so $|V(H)| \geq 13$. $\square$

The Folkman floor and the empty window. The minimum order of a K_4 -free 6-chromatic graph is the vertex Folkman number $F_v(2, 2, 2, 2, 2; 4)$. The literature settles this at

$$F_v(2, 2, 2, 2, 2; 4) = 16$$

(Lathrop and Radziszowski [7]). Its only 16-vertex witnesses are the two Ramsey $(4, 4)$ -graphs (no K_4 , no independent 4-set), which are dense: as a complementary pair they satisfy $m + \bar{m} = \binom{16}{2} =$

120, about 60 edges each (average degree ~ 7.5). Both violate the codegree ceiling, since (1) gives exactly 48 at $n = 16$ and $60 > 48$. Therefore the UDG-necessary class, that is, the class of graphs that are simultaneously K_4 -free and $K_{2,3}$ -free, has no member at the Folkman floor. Because the ceiling (1) only reaches average degree 7.5 (the density of the Ramsey $(4, 4)$ -graphs) around $n \approx 26$, even Folkman-floor density excludes the UDG-necessary class below roughly $n \approx 26$.

Novelty. We stress that the ingredients here are not new. The $K_{2,3}$ -free fact is classical (it is the combinatorial heart of the unit-distance incidence bound [13]), the Kostochka-Yancey bound is a published theorem, and the Folkman number is a literature value. The contribution of this section is the *synthesis*: pinning the missing $\chi \geq 6$ object simultaneously below by criticality (Kostochka-Yancey), above by planarity-of-incidence ($K_{2,3}$ -freeness), and at the abstract floor by Folkman, to conclude that the object cannot be a small dense host.

7 The class contains no 6-chromatic member through order 17

Section 6 pins the missing object inside the UDG-necessary class of simultaneously K_4 -free and $K_{2,3}$ -free graphs. It is natural to ask whether that class contains *any* 6-chromatic member at small order, UDG-realizable or not. Local searches (reported in Section 8) never find one, but a heuristic miss is not evidence of absence. At small orders the question can be settled outright, and this section records the exhaustive answer.

Theorem 2. *The class of K_4 -free, $K_{2,3}$ -free graphs contains no member with $\chi \geq 6$ on at most 17 vertices. Hence the smallest such graph, if one exists, has order at least 18.*

The proof is a computer proof in two halves, sharing all four search-space reductions below but differing in how the resulting search space is exhausted. Through order 16 the space is small enough to enumerate outright, and we do: every candidate is generated and colored. At order 17 enumeration is no longer affordable (see the timings below), so we do not enumerate. Instead we hand the whole question, both quantifiers at once, to a single satisfiability solver whose unsatisfiability proof *is* the nonexistence statement. The two halves are independent, and where they overlap they agree.

Both halves are made feasible by four search-space reductions with short proofs. It suffices to look for a 6-critical member of the class: any $\chi \geq 6$ member contains a 6-critical subgraph, and both defining properties are closed under taking subgraphs, so the critical subgraph lies in the class at some order no larger. The reductions then constrain that critical subgraph: (i) a 6-critical graph is 2-connected and has minimum degree at least 5 (standard critical-graph theory [4]); (ii) it satisfies the Kostochka-Yancey floor $m \geq 2.8n - 1.8$ [6]; (iii) as a member of the class it satisfies the codegree ceiling (1), so its edge count is confined to a narrow per-order window (at $n = 16$: $43 \leq m \leq 48$; at $n = 17$: $46 \leq m \leq 52$); and (iv) it obeys the following maximum-degree cap.

Lemma 2 (Maximum-degree cap). *Every $K_{2,3}$ -free graph on n vertices with minimum degree at least 5 has maximum degree at most $(n - 1)/2$.*

Proof. Fix a vertex a and count the edges that leave its neighborhood, grouping them by their far endpoint. Each neighbor w of a sends $\deg w - 1 \geq 4$ edges to vertices other than a ; on the other side, a vertex $x \neq a$ receives one such edge for each common neighbor of a and x , of which there are at most 2 by $K_{2,3}$ -freeness. Hence

$$4 \deg a \leq \sum_{w \sim a} (\deg w - 1) = \sum_{x \neq a} |N(a) \cap N(x)| \leq 2(n - 1),$$

and $\deg a \leq (n - 1)/2$. □

Proof of Theorem 2, orders $n \leq 16$. By the reductions above, it suffices to enumerate, for each $n \leq 16$, all 2-connected K_4 -free, $K_{2,3}$ -free graphs with minimum degree at least 5, maximum degree at most $\lfloor (n-1)/2 \rfloor$, and edge count in the window $\lceil 2.8n - 1.8 \rceil \leq m \leq \lfloor n(1 + \sqrt{8n-7})/4 \rfloor$, and to check each for 5-colorability: any $\chi \geq 6$ member of the class on at most 16 vertices would contain a 6-critical subgraph appearing in this enumeration at its own order, and that subgraph would not be 5-colorable. The window is empty for $n \leq 12$ (the floor-ceiling comparison in the proof of Proposition 1), and the Folkman number $F_v(2, 2, 2, 2, 2; 4) = 16$ [7] already rules out $n \leq 15$; the orders $n = 13, 14, 15$ were nonetheless enumerated in full as consistency checks.

The enumeration was carried out with the **geng** generator of the **nauty** package [9], extended by a custom pruning plugin that rejects any partial graph containing a K_4 or a $K_{2,3}$. The prune is sound and complete because both forbidden subgraphs persist under **geng**'s vertex-augmentation order: once present in a partial graph they are present in every completion, so rejecting at first appearance loses no member of the class. (A further counting-based prune, a cherry-budget inequality derived from the same double count as (1), accelerates the search.) Table 2 summarizes the outcome. The orders 13 and 14 contain no class member in the window at all; order 15 contains 11, all 5-colorable; order 16 contains 11,315, all 5-colorable (11,291 colored directly by DSATUR; the remaining 24 decided satisfiable by a complete SAT solver, and each independently reconfirmed by the solvers CaDiCaL, Glucose, and MiniSat). Every enumerated graph is 5-colorable, so no 6-critical member of the class exists on at most 16 vertices, and hence no $\chi \geq 6$ member either. $\square$

At order 17 the same strategy is out of reach. The class grows by roughly two orders of magnitude per vertex at this scale, and a direct measurement on the hardware used here put the order-17 enumeration beyond 80 core-days, with the dominant edge-window cell alone holding more than 1.1 million isomorphism classes. What makes that cost avoidable is that the question being asked is not “list the class” but “does the class contain a graph that is not 5-colorable”, and the second question is a two-quantifier statement whose quantifiers need not be separated:

$$\exists G \in \mathcal{C}_n \ \forall c : V(G) \rightarrow [5] \ c \text{ is not a proper coloring of } G.$$

Enumerate-then-color answers it by splitting the quantifiers, paying once per member of $\mathcal{C}_n$. The following proof does not split them.

Proof of Theorem 2, order $n = 17$. Fix an edge count m in the window $46 \leq m \leq 52$ and encode membership in the reduced search space as a propositional formula $\Phi_{n,m}$ over the $\binom{n}{2}$ edge variables of a labeled graph on n vertices: K_4 -freeness and the codegree bound as forbidden-configuration clauses, minimum degree 5, maximum degree $\lfloor (n-1)/2 \rfloor = 8$, and edge count exactly m through reified cardinality counters. A satisfying assignment of $\Phi_{n,m}$ is exactly a labeled graph meeting all four reductions at edge count m .

Solve $\Phi_{n,m}$ with a conflict-driven clause-learning solver carrying two propagators. The first is the isomorphism propagator of *SAT modulo symmetries* [5], which prunes any partial assignment extendable only to lexicographically non-minimal adjacency matrices; since it removes only non-minimal labelings, at least one labeling of every isomorphism class survives. The second is a chromatic-number propagator: whenever the solver reaches a total assignment, the propagator searches for a proper 5-coloring c of the induced graph, and on finding one returns the clause

$$\bigvee_{\substack{uv \notin E(G) \\ c(u)=c(v)}} x_{uv},$$

asserting that at least one currently monochromatic non-edge must become an edge. Every graph falsifying that clause admits c as a proper 5-coloring, so the clause is sound; and it is violated by the

present assignment, so the solver makes progress. Consequently a model of the augmented formula is a member of the class at order n and edge count m that is not 5-colorable, and unsatisfiability of the augmented formula is a proof that no such member exists.

For $n = 17$ every cell $m = 46, \dots, 52$ is unsatisfiable, in 2,975 core-seconds in total (1,202 seconds of wall time on seven cores), the per-cell times running from 13 seconds at $m = 52$ to 1,200 seconds at $m = 47$. The two extremal cells $m = 51$ and $m = 52$ are in fact empty, as the non-vacuity control below records, so their unsatisfiability is immediate and the substance of the sweep lies in $m = 46, \dots, 50$. No graph is emitted at any point. Hence no 6-critical member of the class exists on 17 vertices, and with the previous half of the proof, none on at most 17 vertices; so the class has no $\chi \geq 6$ member on at most 17 vertices. $\square$

The same procedure reproduces the enumerated orders: the order-15 window closes in 3.6 seconds and the order-16 window in 74.3 seconds, against the 66.3 core-hours the order-16 enumeration took. Those two agreements are the reason we report the order-17 result with the same confidence as the enumerated ones: the procedure is not merely faster on the order where we cannot check it, it is faster on the orders where we can, and there it returns the answer we already knew.

Table 2: The UDG-necessary class (K_4 -free and $K_{2,3}$ -free) over the per-order edge window from the Kostochka-Yancey floor to the codegree ceiling (1), restricted to 2-connected graphs with minimum degree ≥ 5 and maximum degree $\leq \lfloor (n-1)/2 \rfloor$. Orders 13 through 16 are enumerated and every enumerated graph is 5-colorable. Order 17 is decided without enumeration, one unsatisfiability proof per cell of the window, so no member is ever generated and the enumeration columns do not apply.

n	Edge window	Class members	By DSATUR	By SAT	$\chi \geq 6$
13	[35, 35]	0	0	0	0
14	[38, 39]	0	0	0	0
15	[41, 43]	11	9	2	0
16	[43, 48]	11,315	11,291	24	0
17	[46, 52]	not enumerated (decided directly)			0

Calibration. The exhaustiveness claim is only as good as the generator and the prune, so both were calibrated against independent implementations. At $n = 7, 8, 9$ the pruned generator’s output was compared with an unpruned enumeration passed through a separately written membership filter (disjoint code): the counts of K_4 -free, $K_{2,3}$ -free graphs agree exactly (352, 2,001, and 15,481), and at $n = 9$ the two canonically labeled output *sets* are identical. At the extremal cell $n = 16, m = 48$, where the codegree ceiling (1) is met with equality, the enumeration emits exactly one graph: the Shrikhande strongly regular graph $\text{srg}(16, 6, 2, 2)$, in which every pair of vertices has exactly 2 common neighbors; it has $\chi = 4$ and is correctly passed over. Finally, the sweep reproduces the Folkman floor of Lathrop and Radziszowski [7] from inside the smaller class: no 6-chromatic K_4 -free graph appears below order 16. Beyond calibration, the headline counts and colorings were re-derived after the fact under independent code paths (a second membership filter, fresh SAT encodings under a different solver, re-enumeration of sampled parts of the order-16 run), and the pruning plugin was fuzz-tested without ever cutting a valid completion.

Calibration of the order-17 procedure. The order-17 half rests on a solver returning unsatisfiable, so the propagator that supplies the coloring half was calibrated against known answers before

it was used, at each of the three chromatic thresholds it could be asked about. At threshold 4: with triangles forbidden, the procedure is unsatisfiable at order 10 and satisfiable at order 11, and the graph it returns at order 11 is isomorphic to the Grötzsch graph, which by Chvátal’s theorem [1] is the unique triangle-free 4-chromatic graph of that order. At threshold 5: with K_4 forbidden, unsatisfiable at order 10 and satisfiable at order 11, agreeing with the minimum order of a K_4 -free 5-chromatic graph [3]. At threshold 6, the value actually used: with K_6 forbidden, unsatisfiable at order 7 and satisfiable at order 8, as it must be, since a k -critical graph has order k or at least $k + 2$ but never $k + 1$, while $C_5 + K_3$ is 6-chromatic of order 8 with clique number 5; the returned graph is isomorphic to $C_5 + K_3$. Every satisfiable answer above was re-verified by an independent complete solver on a separately generated encoding. Two further controls run on the production cells themselves. First, non-vacuity: an unsatisfiable answer is uninformative if the cell is empty for reasons unrelated to coloring, so every cell of the order-17 window was also solved at threshold 3, and the returned graph was passed through the same independent membership filter used above. Five of the seven cells, $m = 46, \dots, 50$, return a graph the filter certifies as a class member, so their threshold-6 unsatisfiability is a statement with content. The two extremal cells $m = 51$ and $m = 52$ are empty. Re-solved with no chromatic constraint at all they enumerate zero graphs, and a disjoint procedure, static lexicographic-leader symmetry breaking under an ordinary CDCL solver sharing no code with the isomorphism propagator, independently reports the same; their unsatisfiability at threshold 6 is therefore vacuous. This costs Theorem 2 nothing, since an empty cell contains no member that could be 6-chromatic, but it does locate the entire content of the order-17 window in $m = 46, \dots, 50$, and controls that bear on the theorem belong there rather than on an extremal cell. Second, ablation: rerunning a cell with the class constraints removed but the order, degree, and edge-count constraints kept must return satisfiable, confirming that the unsatisfiable verdict is a statement about the class rather than a failure of the search. At $n = 17$, $m = 49$ it does, in 221 seconds, and the graph it returns is independently confirmed to have $\chi \geq 6$; it carries a K_4 and a codegree of 5, so it lies outside the class, which is what the ablation intends. We site this control on $m = 49$ rather than on an extremal cell because $m = 49$ is nonempty and the extremal cells are not: an ablation on an empty cell sets one vacuous unsatisfiability against another and establishes nothing.

The intermediate ablations at that cell are worth recording, because they localize the obstruction inside the class. Removing K_4 -freeness alone leaves the cell unsatisfiable, and no more expensive (77 seconds against 171). Removing the codegree bound alone, keeping K_4 -freeness, leaves the cell undecided after 1,800 seconds. At this order, then, it is $K_{2,3}$ -freeness and not K_4 -freeness that forces 5-colorability, which is consistent with the role the codegree bound plays throughout Section 6: the constraint that comes from planar unit-distance geometry is also the constraint that does the combinatorial work.

A second enumerator at orders 15 and 16. Enumeration completeness at the two orders that carry content was independently re-derived with a generator built on different principles. Rather than canonical augmentation, the second enumerator solves the same cell formulas $\Phi_{n,m}$ described below with the isomorphism propagator of [5], emitting one graph per isomorphism class and terminating in unsatisfiability once the class is exhausted; canonicity is enforced inside the solver rather than by the generation order. At $n = 15$ it returns the same 11 graphs as **geng**, graph for graph. At $n = 16$ it returns 11,315 graphs, every cell of the window proved exhausted, and matching the **geng** output graph for graph: 11,315 matched, none present in only one of the two lists, and no property violations. The two enumerations are therefore in exact agreement at both orders.

A second encoding at order 17. The nonempty order-17 cells were also decided a second time under a structurally different formulation. The question $\exists G \forall c$ is natively a 2-QBF, and writing it as one moves the universal half inside the formula: the existential block carries the edge variables and the encoding’s auxiliaries, the universal block a color assignment, and the matrix asserts class membership together with the requirement that the assignment either fail to be a well-formed color assignment or leave some edge monochromatic. The solver then reasons about colorability rather than sampling it one counterexample at a time, so this route shares neither the coloring propagator nor its clause-learning loop with the procedure above. Calibrated first on the enumerated orders, where it returns unsatisfiable on every cell of the order-15 and order-16 windows, it reproduces unsatisfiable on all five nonempty cells of the order-17 window, $m = 46, \dots, 50$, in 45 to 579 seconds each. This is an independent reproduction of the verdict rather than a certificate: the quantified formula is refuted by a solver, and no machine-checkable proof object is extracted from it.

Epistemic status. Three caveats, stated plainly. First, the order-17 half of Theorem 2 rests on unsatisfiability answers from a solver carrying two propagators, and we did not extract machine-checkable proof certificates for them; what stands behind them instead is the calibration above, the agreement of the same procedure with the independently enumerated orders 15 and 16, and the non-vacuity and ablation controls. It is worth being exact about what a certificate here could even mean, because the obvious answer is wrong. One cannot certify these answers with a DRAT or LRAT checker against $\Phi_{n,m}$, and not for want of tooling: $\Phi_{n,m}$ is *satisfiable*: its models are precisely the class members at that order and edge count, which is what the non-vacuity control exhibits. The unsatisfiability being reported is that of $\Phi_{n,m}$ together with a semantic side condition that is not part of the formula, so no propositional refutation of $\Phi_{n,m}$ exists to be checked. (We confirmed this concretely: the solver emits an LRAT file, an independent checker rejects it, and a plain solver run on $\Phi_{n,m}$ without the propagators returns a model.) What a certificate must instead do is discharge the two clause families separately: each coloring clause is justified by exhibiting the 5-coloring that generated it, which is checkable in $O(n^2)$ by confirming that the pairs it lists are exactly the monochromatic non-adjacent pairs of that coloring; and the minimality clauses are justified once and for all by the lexicographic-minimality theorem of [5] rather than per run. A fully certified version therefore wants either that hybrid format or a reformulation as a 2-QBF, $\exists G \forall c$, which is the natural shape of the question and which the solver also supports. We carried out that reformulation, and it reproduces the order-17 verdicts on every nonempty cell, but reproducing a verdict is not certifying it: extracting a checkable proof object from the quantified formula remains open. Second, the two counting lemmas behind the search-space reductions (Lemma 2 and the cherry-budget inequality) are elementary, formalization-ready double-counting arguments, but they have not yet been machine-verified in a proof assistant. Third, both halves inherit the reductions: if a 6-critical graph could evade one of (i)–(iv), it would evade both halves alike, which is why those four statements carry proofs here rather than citations alone. As with the rest of this note, Theorem 2 bounds the abstract class only: it moves no bound on $\chi(\mathbb{R}^2)$ and says nothing about UDG-realizability.

8 Discussion and the target object

The preceding lines of argument pincer the missing ingredient from three sides.

- It cannot live in the realizable lineage (Sections 4-5): the lineage is vertex-critical and hence forcing-sterile, by the Essential-Pair Lemma.

- It cannot be a small dense Folkman-type host (Section 6): such hosts violate the codegree bound that every planar UDG must satisfy.
- It cannot be an in-class graph on at most 17 vertices (Section 7): through order 16 exhaustive enumeration finds every candidate 5-colorable, and at order 17 the question is decided directly, without enumeration.

What remains is a single sharpened target.

Target object. A 6-chromatic graph in the *UDG-necessary class* (simultaneously K_4 -free and $K_{2,3}$ -free), of order at least 18 (Theorem 2) and heuristically of order at least roughly 26, together with an explicit unit-distance embedding in $\mathbb{R}^2$.

In-class membership (K_4 -free and $K_{2,3}$ -free) is a *necessary* condition for UDG-realizability, not a sufficient one. A graph passing both local obstructions still requires an actual coordinate embedding to move $\chi(\mathbb{R}^2)$. The codegree ceiling, in particular, is a counting bound: the true maximum number of unit distances among n points sits well below it at small n , so a graph that respects (1) need not be realizable.

Honest limitations. We list the caveats explicitly.

1. *The Essential-Pair Lemma is elementary and known.* Its corollary is implicit in standard critical-graph theory [4] and appears explicitly as Theorem 3.17 of Martin [8] (phrased there via “implicit-edges” and “implicit-identities”). We include a proof only for self-containedness and make no priority claim.
2. *The $K_{2,3}$ -free fact is classical.* It is the basis of the Spencer-Szemerédi-Trotter unit-distance edge bound. The novelty of Section 6 is the synthesis with Kostochka-Yancey and the Folkman number, not the ingredients.
3. *The density observation is empirical.* The claim that reachable K_4 -free 6-critical hosts sit at $m/n \approx 4.2$ is an observation over roughly 35 cores from one construction family, not a theorem about all K_4 -free 6-critical graphs. Only the $n \leq 12$ exclusion (Proposition 1), the Folkman-floor exclusion, and the $n \leq 17$ non-existence result (Theorem 2, computer-assisted, with the caveats stated in Section 7) are rigorous.
4. *No bound on $\chi(\mathbb{R}^2)$ moved.* This note is negative and explanatory. It tightens the description of where the missing object can live; it does not produce the object.

Outlook. The results suggest a clear procedural lesson: generate *inside* the UDG-necessary class rather than above it. Manufacturing K_4 -free 6-chromatic graphs and cutting them down fails the codegree bound; the alternative is to enforce K_4 -freeness and $K_{2,3}$ -freeness as hard invariants from the start, seeding from a real $\chi = 5$ UDG (which lies in the class automatically) and adding only codegree-safe and K_4 -safe structure until the chromatic number is forced upward. Whether that class forces $\chi \geq 6$ at some attainable order, or instead caps at $\chi = 5$ (the UDG analogue of $\chi(\mathbb{Q}^2) = 2$), is the open question this framing isolates.

Computational probes of this question approach the target object from three heuristic directions and, at small orders, one exhaustive one. (i) *Growth from below.* A greedy search that grows a graph from the empty graph by adding only codegree-safe and K_4 -safe edges caps at $\chi = 4$ across orders $17 \leq n \leq 64$: it never even reaches $\chi = 5$. (ii) *Annealing.* A local search with both edge

additions and deletions (simulated annealing over graphs kept K_4 -free and $K_{2,3}$ -free, minimizing the number of proper 5-colorings) over orders $26 \leq n \leq 40$, with on the order of 20,000 restarts, reaches in-class $\chi = 5$ at orders as small as $n = 29$ but never $\chi = 6$, with the edge density plateauing near 84% of the codegree ceiling (1) at every n . (iii) *Repair from above*. The top-down probe of Section 6, which starts from a genuine $\chi = 6$ graph ($M^3(C_5)$) and repairs its codegree violations while holding $\chi = 6$, stalls with the violations intact in the 6-critical core. The three local searches are heuristic, not non-existence results, and they probe orders up to 64. For $n \leq 17$, however, Theorem 2 now closes the question outright: the class simply contains no $\chi \geq 6$ member there. The in-class negative is thus decisive for $n \leq 17$ and heuristic for n up to 64. It is worth recording what the frontier cost before and after, because the change is not one of hardware. Enumeration walls quickly: four sampled parts of a 4096-way split of the $n = 17$ window each exceeded 1,700 cpu-seconds without finishing, putting the full $n = 17$ enumeration above roughly 80 cpu-days on the workstation used here, with per-order cost multipliers of roughly $29\times$ to $100\times$ along the ladder. Deciding the same order without enumerating costs 0.83 core-hours on that same class of machine. The multiplier along the ladder is unchanged; what changed is the constant, because the cost no longer scales with the size of the class. Whether that buys one more order or several is an empirical question we do not prejudge here. Together the probes and the non-existence results triangulate one conclusion: the target object, if it exists, lives at orders that local moves do not reach from any natural seed and that exhaustive enumeration does not reach at all by generation.

Acknowledgments

This is a note from an ongoing computational program on the Hadwiger-Nelson problem. The realizable lineage graphs studied here derive from the de Grey [2] and Polymath16 [11] constructions. The exhaustive forcing census and the criticality scans were carried out on a single workstation using standard complete SAT solvers (CaDiCaL, MapleChrono, and Glucose, invoked through the PySAT toolkit) together with a purpose-built symmetry-breaking colorability solver; the local searches of Section 8 were run under PyPy. The exhaustive enumeration of Section 7 used the **geng** generator of the **nauty** package [9] with a custom pruning plugin, on the same workstation. No external funding supported this work.

References

- [1] Vasek Chvátal. The minimality of the Mycielski graph. In *Graphs and Combinatorics*, volume 406 of *Lecture Notes in Mathematics*, pages 243–246. Springer, 1974.
- [2] Aubrey D. N. J. de Grey. The chromatic number of the plane is at least 5. *Geombinatorics*, 28(1):18–31, 2018. Preprint: arXiv:1804.02385.
- [3] Tommy R. Jensen and Gordon F. Royle. Small graphs with chromatic number 5: a computer search. *Journal of Graph Theory*, 19(1):107–116, 1995.
- [4] Tommy R. Jensen and Bjarne Toft. *Graph Coloring Problems*. Wiley, New York, 1995. Standard compendium of critical-graph theory (Dirac, Gallai).
- [5] Markus Kirchweger and Stefan Szeider. SAT modulo symmetries for graph generation. In *27th International Conference on Principles and Practice of Constraint Programming (CP 2021)*, volume 210 of *LIPICs*, pages 34:1–34:16. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2021. Isomorphism handled by a lexicographic-minimality propagator inside the CDCL solver.

- [6] Alexandr Kostochka and Matthew Yancey. Ore’s conjecture on color-critical graphs is almost true. *Journal of Combinatorial Theory, Series B*, 109:73–101, 2014. Preprint: arXiv:1209.1050.
- [7] Joel Lathrop and Stanisław P. Radziszowski. Computing the Folkman number $f_v(2, 2, 2, 2, 2; 4)$. *Journal of Combinatorial Mathematics and Combinatorial Computing*, 78:119–128, 2011. Establishes $F_v(2, 2, 2, 2, 2; 4) = 16$ (Nenov had shown $12 \leq F_v \leq 16$); the upper-bound witnesses are the two Ramsey $(4, 4)$ -graphs on 16 vertices.
- [8] José Antonio Martín H. Graph-chromatic implicit relations. arXiv:0901.1255 [math.CO], 2009. Theorem 3.17 (“there are no implicit relations in critical graphs”); forced-different and forced-same non-adjacent pairs are termed implicit-edges and implicit-identities. Unrefereed preprint.
- [9] Brendan D. McKay and Adolfo Piperno. Practical graph isomorphism, II. *Journal of Symbolic Computation*, 60:94–112, 2014. The nauty and Traces graph-isomorphism suite; the **geng** generator is distributed with nauty.
- [10] Jaan Parts. Graph minimization, focusing on the example of 5-chromatic unit-distance graphs in the plane. *Geombinatorics*, 29(3):137–166, 2020. Reduces 5-chromatic unit-distance graphs to 510 vertices, continuing the Polymath16 effort.
- [11] Polymath, D. H. J. Polymath16: The chromatic number of the plane. Online collaborative project, Polymath wiki and blog of Dustin G. Mixon and Aubrey de Grey, 2018. Reduced unit-distance graphs with chromatic number at least 5 to roughly 510 vertices.
- [12] Alexander Soifer. *The Mathematical Coloring Book: Mathematics of Coloring and the Colorful Life of its Creators*. Springer, New York, 2009.
- [13] Joel Spencer, Endre Szemerédi, and William T. Trotter. Unit distances in the Euclidean plane. In *Graph Theory and Combinatorics (Cambridge, 1983)*, pages 293–303. Academic Press, London, 1984. Proves the order- $n^{4/3}$ bound on the number of unit distances among n points in the plane.
- [14] D. R. Woodall. Distances realized by sets covering the plane. *Journal of Combinatorial Theory, Series A*, 14:187–200, 1973. Contains the fact that the unit-distance graph on the rationals squared has chromatic number 2.